Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/30/2021
PI Name:	Sanford, Larry Ph.D.		
Project Title:	Impact of Inflight Stress and Sleep Disturbances on Brain Function, Neural Communication, and Inflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per L. Juliette (Ed., 5/8/22)		
Key Personnel Changes/Previous PI:	July 2021 report: Shao-Hui Chuang and Richard Ciavarra are no longer working on the project. July 2020 report: None.		
COI Name (Institution):	Britten, Richard Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
Grant/Contract No.:	80NSSC19K1582		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts may experience several physical and psychological challenges including prolonged exposure to microgravity, potential social isolation (SI), and predicted exposure to ~13 cGy/Yr Space Radiation (SR). Each of these stressors can have deleterious effects on physical and mental health as well as negatively impact mission performance. Each also can disturb sleep, which can further exacerbate the effects of stress. Determining how astronauts respond to the single and compound effects of mission related stressors is crucial for understanding and mitigating factors that could impede mission success. However, examining their effects will be complex because they have different time courses and durations. SR exposure will be episodic whereas microgravity will be constant and SI and sleep disturbances can be temporary or prolonged. Their effects may also be additive or synergistic, and may vary with sex and with individual differences in stress resilience and vulnerability. Therefore, our goal in this project is to determine the single and synergistic effects of inflight stress and SR on mission related biological and functional challenges that astronauts may face. We will determine how order of experience impacts the effects of inflight stressors and determine sex and individual differences in stress resiliency using an established Wistar rat model. We will use established ground-based models of microgravity (hindlimb unloading), SI, and SR. We will use telemetry to record sleep and to assess temperature increases as an index of the stress response. We will use plasma levels of brain derived neurotropic factor to predict individual differences in stress responses. We will assess fear learning and extinction to determine how chronic inflight stressors impact ability to respond to an additional acute stressor. We will analyze food entrained, anticipatory activity to assess potential impairments in temporal perception. We will determine the effects of inflight stress and SR on sensorimotor function and recognition memory using behavioral tests. We will use local field potential recordings in the amygdala, medial prefrontal cortex, and hippocampus to determine whether stressors and SR alter the coordinated neural activity underlying fear learning and extinction, and we will determine whether there are structural changes within connections between regions. We will use chemokine/cytokine arrays and Ingenuity Pathway Analysis to determine the effect of stressors and SR on the immune system. Collectively these studies will assess functional, mechanistic, and immune consequences of several stressors inherent in space exploration. The data they provide will aid in risk assessment, and may help to develop countermeasures based on individual differences in ability to withstand stress and/or targeting immune pathways that may confer protection against the deleterious effects of multiple stressors that astronauts will face.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will increase understanding of how individual differences in resilience and vulnerability factor in determining lasting outcomes of stressful experiences. They will also determine the role that stress-induced alterations in sleep play in mediating the outcomes of stress. The results will have importance for understanding how individuals cope with stress on Earth and on missions in space.
Task Progress:	Our goal in this project is to determine the impact and potential synergistic effects that multiple inflight stressors and exposure to SR can have on biological systems and mission related performance. Outcomes that we are examining include sensorimotor function and balance (adhesive removal task and balance beam performance), recognition memory (novel object recognition task), time perception (anticipatory activity), fear memory and fear memory related neural communication. We are also examining the relationship of altered performance to the effects of inflight stressors on sleep, and how responses to stressors vary in resilient animals that maintain sleep after experiencing stress and vulnerable animals that have reduced sleep after experiencing stress. This is the second year of this project. We have conducted behavioral and sleep studies on one cohort of control rats (n=20) and in one cohort of socially isolated rats (n=21). Behavioral data from these rats suggest that social isolation can produce alterations in some behaviors including altered behavior in the novel object recognition test and performance on the balance beam. We are continuing analyses that will enable us to distinguish resilient and vulnerable phenotypes and that will enable us to assess potential effects of social isolation on electroencephalographically determined sleep. We are assessing circulating brain derived neurotropic factor (BDNF) levels both prior to, and after, irradiation and exposure to inflight stressor analogs. We have analyzed samples collected prior to irradiation in 57 Wistar outbred rats. Blood samples were collected close to the same circadian time yet the range is large, with concentration values ranging from a few pg/ml up to several thousand across animals. These animals were subsequently irradiated (studies are ongoing); thus, we will be able to determine whether BDNF baseline levels are predictive of stress and SR effects, and whether SR produces alterations in circulating BDNF. We will also be able to determine whether BDNF is predictive of resilient and vulnerable sleep phenotypes. Studies in our first cohort of irradiated rats (n=31 irradiated and 10 shams) are just now beginning. These animals have been irradiated with 15 cGy GCRsim and are now surgically implanted and are beginning behavioral studies and studies of telemetrically recorded sleep. In our studies of neural communication, we have conducted studies in 8 controls and 5 social isolation rats which are in the process of being analyzed. In these animals, we have found coordinated, sleep-related, local activity between brain regions (amygdala and hippocampus) in our fear conditioning paradigm. We are continuing to refine our methods for analysis and data presentation. We are also working on examining the correlation between global EEG activity and local regional activity. Additionally, we also have ongoing studies of local field potentials and sleep in 3 additional SI rats and in 12 irradiated (15 cGy GCRsim) rats. We also are beginning studies to determine if coordinated brain activity occurs in association with anticipatory activity.
Bibliography Type:	Description: (Last Updated: 03/16/2026)
Articles in Peer-reviewed Journals	Britten RA, Wellman LL, Sanford LD. "Progressive increase in the complexity and translatability of rodent testing to assess space-radiation induced cognitive impairment." Neurosci Biobehav Rev. 2021 Jul;126:159-74. Review. https://doi.org/10.1016/j.neubiorev.2021.01.027 ; PMID: 33766676 , Jul-2021